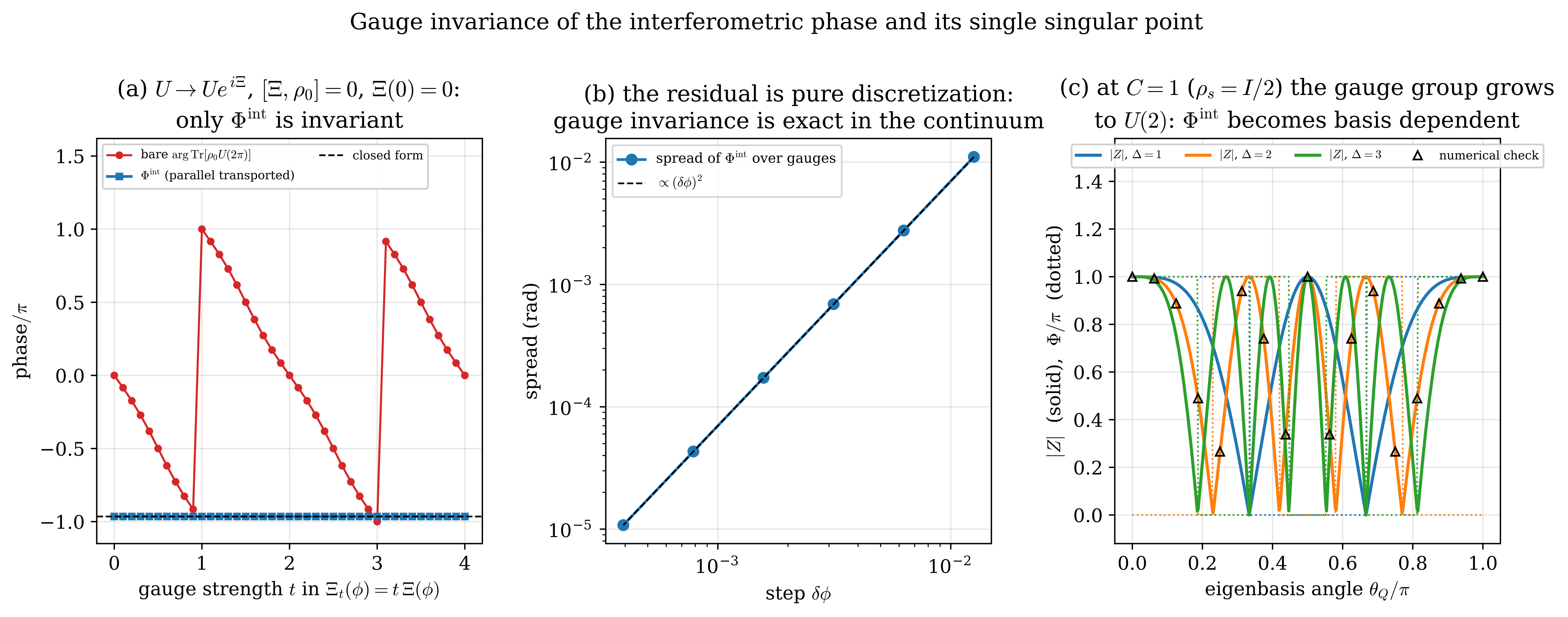

Gauge invariance of the interferometric phase and its single singular point
(a) $U \to Ue^{i\Xi}$, $[\Xi, \rho_0] = 0$, $\Xi(0) = 0$: only $\Phi^{\rm int}$ is invariant
bare $\arg \mathrm{Tr}[\rho_0 U(2\pi)]$
$\Phi^{\rm int}$ (parallel transported)
closed form
phase/$\pi$
gauge strength $t$ in $\Xi_t(\phi) = t\,\Xi(\phi)$
(b) the residual is pure discretization: gauge invariance is exact in the continuum
spread of $\Phi^{\rm int}$ over gauges
$\propto (\delta\phi)^2$
spread (rad)
step $\delta\phi$
(c) at $C = 1$ ($\rho_s = I/2$) the gauge group grows to $U(2)$: $\Phi^{\rm int}$ becomes basis dependent
$|Z|$, $\Delta = 1$
$|Z|$, $\Delta = 2$
$|Z|$, $\Delta = 3$
numerical check
$|Z|$ (solid), $\Phi/\pi$ (dotted)
eigenbasis angle $\theta_Q/\pi$